AutoModel Advisor: Closed-Loop Agentic Model Selection for Cold-Start Datasets

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Abstract

Hugging Face alone hosts over 2.8M pre-trained models, turning "which model should I use for my dataset?" into a search problem that practitioners still solve largely by hand. Prior graph-based approaches such as TransferGraph and ArtifactLinker reframe model selection as link prediction on a model–dataset graph, but they stop at a *predicted* score and never verify it, and they cannot score a dataset that is not already a node in their graph. We present AutoModel Advisor, an agentic system that splits model selection between a Retrieval Agent, which proposes a constraint-aware shortlist for a cold-start dataset by inducing a neighborhood in ArtifactLinker's artifact graph, and an Evaluation Agent, which writes and runs real evaluation code to measure the shortlist *de facto* on the user's data. An orchestrator closes the loop: error analysis from real runs is fed back into the next retrieval round, so recommendations improve with evidence rather than a single blind pass. We describe the two-agent architecture, our evaluation protocol against a raw-GNN baseline and a strong coding-agent baseline, and report preliminary pilot results across ten datasets spanning text, image, and QA tasks.

1 Introduction

Selecting a pre-trained model for a new task typically requires practitioners to use prior knowledge and practical constraints to shortlist candidates, evaluate them on their data, and choose the best-performing model (You et al., 2021; Zhang et al., 2023; Li et al., 2024). With millions of available models (Laufer et al., 2025; Castaño et al., 2023; Ait et al., 2023), however, finding the best candidates remains challenging even under constraints such as parameter budget, license, and modality: the space of candidate model–dataset pairs is enormous, and exhaustively verifying each candidate through actual training or evaluation is computationally intractable (Urbanowicz et al., 2022; Yu et al., 2026).

Recent work reframes model selection as *link prediction* on a heterogeneous model–dataset graph: TransferGraph (Li et al., 2024) trains a regressor over Node2Vec embeddings of a model–dataset graph to predict fine-tuning accuracy, and ArtifactLinker (Yu et al., 2026) trains a joint graph neural network (GNN) with a link-prediction head and an attribute-regression head over an artifact graph of 14,053 models, datasets, papers, and code repositories (51,337 relations), scoring every unobserved (model, dataset) pair as *link probability* $\times$ *predicted attribute value*. Two limitations remain. First, both methods only work for datasets already present as nodes: a genuinely new, cold-start dataset has no edges and is unscorable. Second, both stop at a predicted score; nothing in the pipeline confirms that the top-ranked model actually performs well when it is fine-tuned or evaluated on the target data.

We present **AutoModel Advisor** (Figure 1), an agentic system for model selection on cold-start datasets. Given a target dataset, task, and practical constraints (e.g., maximum parameter count), an orchestrator coordinates two agents. The **Retrieval Agent** first makes the cold-start dataset representable: an LLM summarizes the dataset's card into a structured description (task, domain, size, format), and the summary is embedded with a text embedding model. It then builds on ArtifactLinker with two strategies: (i) *cosine retrieval*, which retrieves the top-K datasets in the artifact graph whose summary embeddings are most similar to the target's, assembles the models measured on those neighbors and their observed scores into *graph-conditioned* evidence, and prompts an LLM to reason over that evidence and recommend candidates; and (ii) *direct GNN scoring*, which inserts the target dataset into the graph as a new node (using its summary embedding as the node feature)

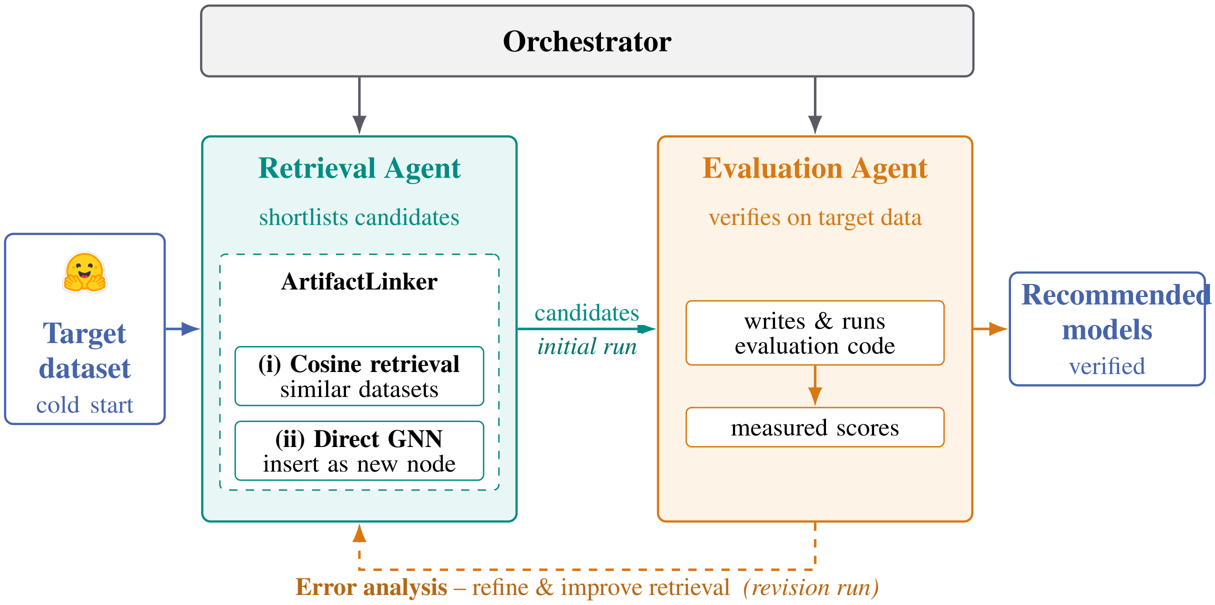

Figure 1: Overview of **AutoModel Advisor**. An orchestrator coordinates two agents on a cold-start target dataset: the Retrieval Agent shortlists candidate models using ArtifactLinker via (i) cosine retrieval over similar datasets or (ii) direct GNN scoring of the dataset as a new node; the Evaluation Agent measures the candidates on the target data. Its error analysis feeds a revision run, closing the loop.

and queries ArtifactLinker's trained GNN to rank every candidate model inductively. The **Evaluation Agent** then writes and executes evaluation code for the shortlisted models on the target data, replacing each predicted score with a measured one. The orchestrator returns these measurements, along with an error analysis of where each model fails, to the Retrieval Agent, which confirms, drops, or replaces candidates for the next round. This closes the retrieval–evaluation loop: recommendations are refined by empirical evidence rather than fixed after a single prediction.

Our contributions are: (1) a two-agent, orchestrated architecture that turns single-shot graph scoring into closed-loop, de-facto-verified model selection; (2) two complementary cold-start strategies that make an out-of-graph dataset scorable by ArtifactLinker's inductive machinery; (3) an evaluation protocol that isolates the contribution of each evidence source (dataset-similarity retrieval vs. GNN prediction) and compares the full agentic system against a raw-GNN baseline and a strong general-purpose coding-agent baseline under matched compute and evaluation-protocol constraints.

2 ArtifactLinker

Our retrieval agent is built on top of ArtifactLinker (Yu et al., 2026), a two-stage *rank-then-verify* framework for automatic state-of-the-art discovery on Hugging Face. ArtifactLinker represents Hugging Face as a heterogeneous artifact graph with model, dataset, paper, and code-repository nodes, connected by edges such as *fine-tune*, *reference*, and *evaluation* (with the measured metric, e.g. F1, as an edge attribute). A shared GNN encoder embeds every node from its neighbors; two heads read the same embedding jointly during training: a link head that estimates $P(\text{compatible})$ for a (model, dataset) pair, and an attribute head that estimates the expected evaluation score $\mathbb{E}[\text{score}]$ if that pair were evaluated. The two are trained together so that the link head, which sees negative (unobserved or incompatible) pairs, teaches the attribute head which pairs are viable to begin with – a signal the attribute head could not learn from observed scores alone. The combined ranking score for an unobserved pair is rank $=$ link $\times$ attribute. ArtifactLinker's second stage lets an LLM coding agent reproduce the actual benchmark for only the top-ranked candidates, verifying predictions instead of trusting them outright.

This reframing – model selection as link prediction on a joint artifact graph, following the same problem framing pioneered by TransferGraph's model-zoo graph learning (Li et al., 2024) – is precisely what our Retrieval Agent reuses as a tool: we load ArtifactLinker's trained checkpoint and query its link $\times$ attribute score for arbitrary (model, dataset) pairs, including a dataset we insert ourselves. What ArtifactLinker does not provide out of the box is (a) a way to place a genuinely new, unseen dataset into the graph, and (b) a mechanism to act on verification failures by retrieving again; both are the subject of Section 3.

3 Our Agentic Approach

Model selection is divided between two cooperating agents, coordinated by an orchestrator that routes work, retries failed steps, and determines when to stop. The Retrieval Agent identifies promising candidates, while the Evaluation Agent verifies their performance on the target data.

Retrieval Agent: making a cold-start dataset scorable. A cold-start dataset has no existing node or evaluation edges in the artifact graph, and therefore cannot be scored directly. To simulate this setting, we use a new Hugging Face dataset that is absent from ArtifactLinker's graph. Both retrieval strategies share two preprocessing steps. First, GPT-4o (OpenAI, 2024) summarizes the dataset card into a structured description of its task, domain, size, and format, using the same prompt employed by ArtifactLinker to summarize its dataset nodes. Second, we embed this summary with Voyage AI's `voyage-3` model (Voyage AI, 2024), matching ArtifactLinker's node features and placing the target dataset in the same representation space as the 1,208 existing dataset nodes. The strategies then diverge in how they use that graph:

Approach 1 – cosine retrieval. We compute cosine similarity between the target dataset and all existing dataset nodes, retrieving the most similar datasets and their associated model evaluations. These models, together with their measured scores and metrics, form a *graph-conditioned* prompt that GPT-5.5 (OpenAI, 2026) uses to reason across similar datasets and recommend the top-5 candidate models, with a justification for each.

Approach 2 – direct GNN scoring. We insert the target dataset into ArtifactLinker's graph as a new node, using its `voyage-3` embedding as the node feature, and apply the trained GNN to rank candidate models by their predicted link probability $\times$ evaluation score. This tests ArtifactLinker inductively, as the target dataset was unseen during training. The top-ranked models and their supporting evaluation evidence are then passed to GPT-5.5 (OpenAI, 2026) to recommend the top-5 candidate models, with a justification for each.

Both approaches are constraint-aware: candidates that violate the user's stated constraints (e.g., parameter budget) are filtered before ranking. The retrieval LLM receives two distinct evidence blocks – ArtifactLinker (GNN) context and dataset-similarity context – as separate, labeled sources, so its recommendation can be traced back to which evidence supported which pick.

Why both? The two approaches fail in different regions of the graph. Cosine retrieval is only as good as the target's nearest neighbors: when the graph contains closely related datasets (e.g., a new sentiment or news-classification set next to many well-evaluated ones), it hands the LLM directly measured scores on near-identical tasks – interpretable, metric-aware evidence that is hard to beat. But it sees only the models evaluated on those few neighbors, and when the target lies in a sparse region (a niche medical or legal task whose closest neighbor is unrelated), that evidence is thin or misleading. Direct GNN scoring instead ranks every model in the graph, aggregating multi-hop signal – a model's results across many datasets, its fine-tuning lineage, shared papers and code – so it can credit models that transfer broadly even when no single neighbor is close. Its weaknesses are the mirror image: the inductive node carries only its text embedding, its scores are opaque regression outputs rather than measured metrics, and the link head favors heavily evaluated, popular models regardless of fit. We therefore expect cosine retrieval to win where the neighborhood is dense and GNN scoring where it is sparse or where the best model's evidence is structurally rather than semantically adjacent; the constraint-aware merge lets the LLM weigh both, and the ablations in Section 4 isolate each signal's contribution.

Evaluation Agent: measuring performance de facto. The Evaluation Agent takes the candidate models and the user's dataset, plans an evaluation, writes and runs the evaluation code, handles execution errors and retries, and records the measured score for each candidate – the same verify step ArtifactLinker applies to its own top-ranked links, now applied to our retrieval agent's cold-start shortlist.

Closed-loop refinement. Model selection proceeds in two runs. In the *initial run*, the Retrieval Agent produces a ranked shortlist and the Evaluation Agent measures each candidate on the user's data. The orchestrator then analyzes where each candidate fails (which classes, slices, or examples it gets wrong) and launches a *revision run*: the Retrieval Agent is re-prompted with the original graph evidence, the initial-run shortlist, and the measured results with error notes, and confirms, drops, or replaces candidates based on what actually happened rather than re-predicting from the graph alone. Selection thus becomes iterative experimentation that targets its own failures instead of a single blind pass.

4 Experimental Setup

Data. We draw candidate query datasets from a pool of 97 Hugging Face datasets (`candidate_datasets_100.txt`), and constrain candidate models to those with at most 14B parameters that appear in ArtifactLinker's graph, so that recommendations are both retrievable and reproducible under a bounded compute budget. For the human-in-the-loop closed-loop evaluation, we select 10 datasets spanning five task families: text classification (`fancyzhx/ag_news`, `stanfordnlp/imdb`, `TimSchopf/medical_abstracts`, `ccdv/arxiv-classification`), claim verification (`ImperialCollegeLondon/health_fact`), summarization (`allenai/scitldr`), reading-comprehension QA (`deepmind/narrativeqa`), image classification (`ethz/food101`, `blanchon/EuroSAT_RGB`), and document VQA (`lmms-lab/DocVQA`), each with a task-appropriate metric (accuracy, macro-F1, ROUGE-L, or ANLS).

Dataset diversity. The 97-dataset candidate pool is text-only by design, so that diversity comes from domain and task structure rather than modality: each entry was chosen to span a distinct knowledge domain (medicine, science, law, finance, literature, general knowledge/commonsense, news, social media, dialogue, fact verification, mathematical reasoning, government/politics, and customer service) while remaining solvable enough for models at or below the 14B constraint to be competitive on it. Task formats are correspondingly heterogeneous rather than uniform: single-label classification and multiple-choice ($\sim$40 datasets, e.g. `cais/mmlu`, `facebook/anli`, `fever/fever`), span-level named-entity and keyphrase extraction ($\sim$10 datasets, e.g. `bigbio/chemprot`, `midas/inspec`), free-text generation spanning extractive/abstractive QA, summarization, and dialogue ($\sim$30 datasets, e.g. `deepset/covid_qa_deepset`, `allenai/mslr2022`, `knkarthick/samsum`), and a small number of continuous-score regression tasks (e.g. `james-burton/wine_reviews`). Every candidate was verified to (i) exist on the Hugging Face Hub and (ii) be absent from ArtifactLinker's evaluation graph – checked via exact, normalized-suffix, and prefix-containment matching against the graph's dataset nodes – to guard against cold-start contamination.

Quantity	Value
Evaluation datasets	10
Conditions per dataset	3 (A/B/C)
Ranked recommendations per condition	5
Total recommendations	150
Unique (dataset, model) runs	97

Table 1: Evaluation-protocol coverage for the closed-loop pilot.

Conditions. For each of the 10 evaluation datasets, we generate top-5 ranked recommendations under three evidence conditions that are identical in every respect (prompt format, LLM, $k = 5$) except which retrieval evidence the recommending LLM sees: **A_merged** (both Approach 1 and Approach 2 evidence), **B_method1_only** (cosine-retrieval evidence alone), and **C_method2_only** (direct-GNN evidence alone). This isolates how much each retrieval signal contributes to the final recommendation. Table 1 summarizes the resulting design: 150 recommendations ($10 \times 3 \times 5$) deduplicate to 97 unique (dataset, model) evaluation runs.

Metrics. For each (dataset, condition), we report **Best@5** (the maximum measured score among its 5 recommended models) and **Mean@5** (averaged over valid ones), plus a **validity rate** – models that cannot be run at all (wrong modality, broken repository, unusable size) are recorded as invalid rather than silently skipped, since validity is itself part of what a good recommender should get right.

Baselines. We compare against two baselines under matched compute and protocol: (1) **raw GNN**, the top-5 candidates by ArtifactLinker's combined link $\times$ attribute score alone, with no LLM reasoning, isolating what the retrieval LLM adds over the graph signal; and (2) a **strong coding-agent baseline**, where a general-purpose LLM coding agent (e.g., Claude) receives the same dataset, task description, and constraints, and is left to search for and evaluate candidate models on its own, without access to our graph-based retrieval tools. Both baselines are run on the same pilot subset, with the same evaluation protocol, hardware, constraints, and compute budget as our full system, isolating whether closed-loop, graph-grounded retrieval outperforms search-and-trial-run alone.

5 Results

We report preliminary results from the pilot described in Section 4; the full human-evaluated comparison across conditions and baselines is an ongoing closed-loop run and we report what is available at submission time. Qualitatively, the retrieval evidence is grounded in measured cross-dataset performance rather than blind LLM guesses: for `ImperialCollegeLondon/health_fact` under condition A, the top-ranked candidate, `bespokelabs/Bespoke-MiniCheck-7B`, is justified by its measured accuracy of 0.774 on the closest retrieved neighbor dataset, `lytang/LLM-AggreFact` – a fact-verification benchmark whose task closely matches health-claim checking – rather than by an unsupported similarity heuristic.

The pilot also surfaces two informative failure modes the closed loop is designed to catch. `blanchon/EuroSAT_RGB` is a "soft" cold start – ArtifactLinker's graph already contains content-variants of EuroSAT, so cosine retrieval finds unusually strong neighbors and should be interpreted separately from genuine cold starts. Conversely, `TimSchopf/medical_abstracts` has the weakest retrieval neighborhood in our pilot (top similarity of 0.67 to an unrelated dataset), making it a useful probe for how gracefully each condition degrades when retrieval evidence is weak; we expect the error-analysis loop (Section 3) to matter most on datasets like this one, where a single retrieval pass has the least evidence to work with.

Quantitatively, we plan to report Best@5 and Mean@5 per condition (A/B/C), the raw-GNN baseline, and the coding-agent baseline, both per dataset and averaged, together with the validity rate per condition, once the real evaluation runs and closed-loop revision round complete. We view the design and instrumentation in Table 1 – specifically that comparing condition C to the raw-GNN baseline isolates what LLM reasoning adds over graph scores alone, and comparing A/B/C isolates each retrieval signal's contribution – as itself a contribution that a full-scale run can populate without redesign.

6 Related Work

Pre-trained model selection. LogME (You et al., 2021) scores transferability of a pre-trained model to a target task from its features alone, without fine-tuning. Model Spider (Zhang et al., 2023) learns to rank a model zoo efficiently using per-model embeddings. TransferGraph (Li et al., 2024) reframes model selection as link prediction on a model–dataset graph, learning a regressor over graph embeddings; it directly inspired the dataset-similarity view we use in our first approach, though we do not integrate its trained predictor. ArtifactLinker (Yu et al., 2026) extends this graph-learning framing to a joint artifact graph spanning models, datasets, papers, and code, and pairs ranking with LLM-agent verification; it is the trained backbone our Retrieval and Evaluation agents both build on. These methods differ from ours in two ways. LogME and Model Spider must run every candidate over the target data, so their cost grows with the model zoo and they ignore the evaluation history the community has already recorded; the graph methods exploit that history but only for datasets that already exist as nodes, and they stop at a predicted score. AutoModel Advisor makes an out-of-graph dataset scorable (by summary-embedding retrieval and inductive node insertion), reasons over the graph's measured evidence with an LLM instead of probing each candidate, and treats the resulting ranking as a hypothesis to verify rather than the answer.

LLM- and agent-based discovery. HuggingGPT (Shen et al., 2023) uses an LLM to plan and dispatch tasks to Hugging Face models but does not rank or verify candidates against measured performance. AutoML-Agent (Trirat et al., 2024) orchestrates a multi-agent LLM pipeline across the full AutoML lifecycle, from data preparation to deployment. In both, the choice of model rests on the LLM's parametric knowledge of model cards and popularity. AutoModel Advisor narrows the scope to model selection but grounds the LLM's choice in graph-conditioned evidence – measured scores on similar datasets and GNN predictions – and filters by user constraints before ranking; our coding-agent baseline, which is given the same candidate pool but no graph evidence, approximates this evidence-free setting (see Limitations).

Closed-loop experimentation. DS-Agent (Guo et al., 2024) augments LLM-driven data science with case-based reasoning over past runs. R&D-Agent (Yang et al., 2025) automates iterative research-and-development cycles by feeding evaluation feedback into subsequent iterations. AutoModel Advisor shares this closed-loop spirit but applies it specifically to model *selection*: rather than iterating on a solution's code or pipeline, the loop iterates on the candidate set, feeding per-model measured results and error analysis (Section 5) back into retrieval so the revision run confirms, drops, or replaces candidates based on what actually happened on the target data.

7 Conclusions

We presented AutoModel Advisor, an agentic system that turns single-shot, predicted model selection into a closed loop of retrieval, real evaluation, and error-driven refinement, and that makes cold-start datasets scorable by ArtifactLinker's graph-based backbone through two complementary strategies – cosine-similarity neighborhood induction and direct inductive GNN scoring. Our evaluation protocol isolates the contribution of each retrieval signal against a raw-GNN baseline and a strong coding-agent baseline under matched constraints. Preliminary pilot evidence shows the system grounding its recommendations in measured cross-dataset performance and surfaces concrete failure modes – soft cold starts and weak retrieval neighborhoods – that the closed loop is designed to address. Completing the full human-evaluated comparison across conditions and baselines, and running additional closed-loop rounds, is ongoing work.

Limitations

Text-only datasets. Our evaluation datasets are text-only by design (Section 4), so that diversity comes from domain and task rather than modality and so that small models remain competitive under the parameter budget. This isolates domain generalization but leaves open how the approach transfers to image, video, audio, and multimodal datasets. Nothing in the pipeline is text-specific – the dataset card is summarized and embedded regardless of modality, and ArtifactLinker's graph already contains vision and speech artifacts – but the neighborhoods are sparser there, evaluation metrics are more heterogeneous, and the Evaluation Agent would need to handle modality-specific preprocessing. Whether the relative strengths of cosine retrieval and direct GNN scoring hold in these regions of the graph is an open question.

Closed candidate pool. Both retrieval approaches, and the coding-agent baseline, choose from the 9,848 models present in ArtifactLinker's graph (5,830 of which satisfy the 14B-parameter constraint used in our runs). This closed list is what makes the comparison controlled and the GNN applicable, but it excludes models released after the graph snapshot and models that were never evaluated on a Hugging Face dataset, which are often the ones a practitioner most wants to know about. Lifting this restriction is not simply a matter of removing the list: with an unconstrained pool, every method degenerates to recommending the strongest available model, so meaningful comparison still requires practical constraints (parameter budget, license, inference cost) that keep the choice non-trivial. Extending the graph incrementally with newly released models, and evaluating the system on an open pool under such constraints, is a natural next step.

Cold-start for the graph, not for the LLM. We define a cold-start dataset as one absent from ArtifactLinker's graph, but the datasets we evaluate on are public Hugging Face benchmarks, and the LLMs in the loop – GPT-4o for summarization, GPT-5.5 for recommendation, and the coding-agent baseline – have almost certainly seen these datasets, their leaderboards, and papers evaluating models on them during pretraining. Part of what the recommender does may therefore be recall rather than reasoning over the retrieved evidence, which inflates every LLM-in-the-loop condition relative to the raw-GNN baseline. What we measure is generalization to datasets *outside the graph*, not to datasets outside the LLM's training distribution; the setting a practitioner most cares about – private or newly collected data – remains untested. A held-out set of datasets released after the LLMs' training cutoffs would separate the two, and we leave this to future work.

Baseline confounds evidence with model. Our coding-agent baseline runs Claude Opus 4.7, whereas the retrieval conditions use GPT-5.5. A gap between the baseline and the merged condition therefore cannot be attributed to graph evidence alone, since it may partly reflect differences between the two models. The clean ablation is the same LLM, prompt, and candidate list with the evidence blocks removed; the baseline should be read as a strong practical reference point – what a practitioner gets today by asking a coding agent – rather than as a controlled measurement of the value of graph evidence.

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